

Semi-Automatic Semantic Annotation of Images

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Abstract

Detailed, consistent semantic annotation of large collections of multimedia data is difficult and time-consuming. In domains such as eScience, digital curation and industrial monitoring, fine-grained high-quality labeling of regions enables advanced semantic querying, analysis and aggregation and supports collaborative research. Manual annotation is inefficient and too subjective to be a viable solution. Automatic solutions are often highly domain or application specific, require large volumes of annotated training corpora and, if using a ‘black box’ approach, add little to the overall scientific knowledge. This article evaluates the use of simple artificial neural networks to semantically annotate micrographs and discusses the generic process chain necessary for semi-automatic semantic annotation of images.

1. Introduction

Semantic annotation of media is recording high-level descriptive terms about the content of the media. It may be coarse-grained (descriptions at the image level) or fine-grained (descriptions at the segment or region level). Manual annotation (i.e., annotation by a human expert) is expensive, time-consuming, inconsistent and subjective. Tools and algorithms are available that can automatically extract low-level feature data from media objects such as color, shape, size, trajectory etc. However, these features are insufficient to support the queries required by domain experts who prefer to access data using higher-level terms such as catalyst, gaseous microemboli or mitochondria. Bridging the distance (often called the “multimedia semantic gap”) between automatically extracted low-level features and high-level semantic terms is the focus of a great deal of research.

The concept of the “semantic gap” was initially defined in psychology [1] and refers to the distance between information that can be extracted from the visual data and the interpretation that different users have for this same data. The difficulty is that humans cannot always describe what they see or explain why they interpret it in a certain way. Also there is rarely complete agreement between a group of users about the interpretation of media content. Because of this, choosing the algorithms to extract low-level information from the media is also more complicated as we do not know what the key features are and therefore what procedure to apply.

Overcoming or mitigating this semantic gap to enable rapid and accurate semantic annotation of images is of particular importance in domains such as biology, geology, astronomy and industrial monitoring. Fields like these use high-resolution, high-throughput sensors and analytical machines to produce very large volumes of media content. Researchers need to be able to analyze and label objects of interest within these images and manage the quantity of data they provide.

This article discusses an approach to addressing the semantic gap using artificial neural networks in the context of the requirements for semi-automatic semantic annotation of scientific media. It summarizes of some of the main approaches to bridging the semantic gap (section 2) and outlines the motivations and characteristics of a sample application (section 3). We present a generic processing chain for classifying images (section 4) and discuss the key challenges. Section 5 describes the use of artificial neural networks to classify image regions of a 3D slice sequence and presents the results of an initial evaluation.

2. Related Work

A variety of research efforts have investigated the use of techniques to extract semantic labels from low-

level features. This section presents some examples of the more common approaches.

Interfaces that assist users to annotate or to link semantic terms to examples or sets of visual features enable expert users to define mappings from features to semantic terms [2,3,4].

More traditional approaches [5–11] have used machine-learning techniques such as statistical analysis, hidden markov models or artificial neural networks to determine semantic terms based on sets of low-level features such as color histograms, texture or shape. These approaches often use a selected set of ‘examples’ and then apply an indexing technique (e.g., latent semantic indexing [6], statistical models [7], statistical factor graphs [8,9], heuristic techniques [10], Minimum Description Length [11]) to develop a model or classification function and label or classify media. These type of statistically-based methods, generally require a significant training set and are unable to incrementally learn as new data is generated and added to the dataset.

These approaches can also be augmented by the use of taxonomies or ontologies [12–15], either in combination with a user interface or with machine-learning approaches. The use of more structured and formalized terminologies enables reasoning using the relationships defined in the ontology and query expansion for better searching over the annotated dataset. This can lead to more flexible, interoperable systems that are easier to extend and explain.

Previous work by Perner has shown that neural networks to tend generalize better than other methods [16] and can model non-linear decision surfaces. Colantonio et al. [17] and Di Bona et al. [18] use neural networks to segment and characterize medical images. However, neural networks also require a labeled training set of suitably significant size and variation. In contrast, a decision tree is a method that can easily be trained but they do not generalize as well as neural nets. If the variation in the data is very high then the preferred method would be case based reasoning [19,20]. This method does not necessarily generalize; it relies on samples and can incrementally learn. Unlike neural networks, both decision trees and case-based reasoning have the capability to explain their classifications.

The commonalities between these different approaches include the requirement for appropriate extraction of sets of low-level media features and the need for high-level, preferably well-defined, semantic terms to label and classify the media. The approaches that employ machine-learning techniques usually require a significant set of examples for training. The next section of this article presents an example

application that needs support for automatic or semi-automatic annotation of image regions but is not able to initially supply a training set of statistically significant size.

3. Application

At the Institute for Molecular Bioscience of the University of Queensland, the Visible Cell project [21,22] aims to increase understanding of the mammalian cell via the synthesis of physical data, models, mathematical and statistical simulations, and bioinformatics data. The objective of the project is to provide a visualization environment that seamlessly embeds macro-molecular structures, networks and quantitative simulations based on mathematical and complex-system models into a 3D mammalian cell reconstructed from high resolution tomograms and electron micrographs.

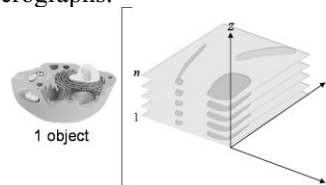


Figure 1: 2D slices of a 3D object

Figure 1 (3D cell structure adapted from http://commons.wikimedia.org/wiki/Image:Biological_cell.svg) illustrates the data in this application which consists of 2D micrographic digital images of thin slices taken sequentially through a cell from a pancreas (a 3D object). We had 31 digital images from one cell. These 2D slices are used to create a 3D model of the sub-cellular structure. To build this 3D model and support the type of integrated, semantic queries desired by users, we need to label the single sub-cellular objects within the 2D slices. The objects of interest in this project are: endoplasmic reticulum (ER), golgi (Go), mature granules (MG), mitochondria (Mi), ribosome (Ri) and tubular vesicles (TV).

A sample image is shown in Section 4.2, Figure 2. Finding automatic image segmentation algorithms for images like this one is not easy since the cell is highly structured. Parallel research at the IMB [23] is working on algorithms for automated segmentation. However, until efficient and accurate automatic segmentation algorithms are able to be developed the method of choice is to manually label the objects. This means that a human user is sitting in front of a computer and circles objects of interest they can detect. The correct label for each object is not always obvious to the user and can only be determined after examining other images in the 3D stack to find adjacent regions that could be more easily identified. The user was able to

label 7580 regions in the set of 31 slices. This is only the regions that the user was able to manually circle and identify – we cannot be certain that the user was able to identify all possible objects correctly.

To correctly label all these regions is time-consuming and difficult for a human user. Therefore an approach to automate part or all of the procedure is necessary. Initially, work using semantic inferencing rules was extended to evaluate the Rules-By-Example interface [3,25]. In this article we use artificial neural networks (ANN) since they are widely used in image classification [15] to discuss the generic process for semantic annotation and classification of images.

Our dataset consists of 5548 entries. Of the 7580 regions labeled by the expert, 2032 were identified using class labels that were not used in this evaluation being of minor objects that were of less interest. Each entry represents a region with the set of extracted low-level features and the user-assigned label. The class distribution in the dataset is: ER–2486; Go–1463; Mi–221; MG–105; Ri–1125; TV–148. The evaluation of the network, in terms of the accuracy rate, was done by test and train. The dataset was divided into sets of 70% for training and 30% for testing. The neural network architecture used was a simple feedforward network trained using a backpropagation algorithm.

The features in the dataset are: Area, ConvexArea, FilledArea, MajorAxisLength, MinorAxisLength, Eccentricity, Solidity, Extent and DominantColor. Excluding DominantColor, which was calculated separately, these terms are extracted using MATLAB's 'regionprops' function from the Image Processing Toolbox. Through discussion with the expert users, general features such as size and shape were noted as being of particular importance and usefulness in distinguishing the different objects in the cell micrograph. This process of knowledge acquisition is not an easy procedure and requires experience to be able to extract relevant and useful knowledge from expert users. A methodology for doing so is presented in [26].

4. (Semi-)Automatic Annotation

The previous section presented an example application that would benefit from the ability to semi-automatically semantically annotate images. This section discusses the different levels of “semantic” labels and then describes the generic semantic annotation process for images.

4.1. Semantic Labels

Figure 2 shows the different types of semantic labels that can be applied to an image and gives some examples of the features and some possible values. The automatically or semi-automatically extracted *low-level* features have numerical values and, as such, are not easily understood or interpreted by a human user.

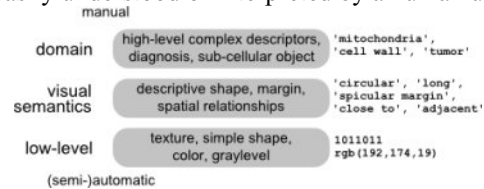


Figure 2: Overview of different levels of semantic annotations

Of more use for human interpretation are *visually descriptive* or *semantic features* which describe characteristics in more usable, often symbolic vocabularies and may be drawn from standards such as BIRADs® (Breast Imaging Reporting and Data System, American College of Radiology). The highest-level of semantic labels are *domain specific* descriptors. These may be defined in a domain ontology, taxonomy or standard such as MeSH (Medical Subject Headings, National Library of Medicine) or the Gene Ontology [27] and are rich, descriptive terms about the content of the image.

The visual semantic terms can be used to define mappings to the domain level terms. For example, “if object is long and thin and close to an object that is identified as ‘Golgi’ then the object is a ‘Golgi’ ”.

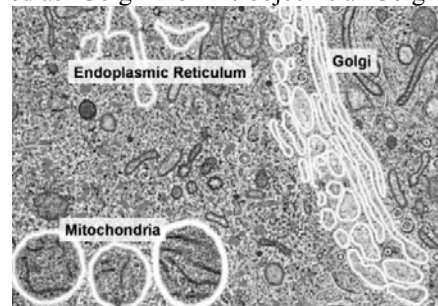


Figure 3: Sample segmented pancreas cell micrograph, example regions for Endoplasmic Reticulum, Golgi and Mitochondria have been highlighted.

In the example application described in section 3, the 2D slices contain objects that are described by human experts using both visual semantic terms and labeled or classified using domain terms. For example, the sections labeled ‘Mitochondria’ in Figure 3 have an uneven texture with distinct internal striations; they are circular in shape and generally large in area. In contrast the regions which make up the structure labeled ‘Golgi’ have less distinctive visual features in common; they are generally smooth in texture, often

long and thin in shape. Their most distinctive visual characteristic is their spatial relationships as the regions tend to lie in long, parallel alignments close to each other.

As you can see these descriptions are not numerical values but rather intermediate descriptors used to describe the objects of interest in terms of their shape, texture and location within the complete scene. In order to generate the high-level domain terms, useful for querying and to assist in generating the final 3D models, semantic mappings need to be developed from the low-level to the intermediate terms and then from the intermediate to the domain level descriptions. The next section describes a generic process chain for image annotation and classification.

4.2. The Generic Process Chain Necessary for Semi-Automatic Semantic Annotation

The process of semantically annotating images based on low-level features usually follows a common abstract procedure. The generic processing chain for image understanding is shown in Figure 4.



Figure 4: Generic process chain for image annotation or classification

The first step is segmentation where the image is segmented into background and objects of interest. Ideally algorithms for automatic segmentation should be used to analyze and divide the images. However, devising a general procedure for segmenting images is not always possible and sometimes manual or semi-automatic processes are required.

Once we have determined what image pixels belong to which object, we need to label the regions representing the objects. Using these regions, we can then extract the object features (e.g., low-level features such as graylevel, simple shape features, texture and color) to produce a feature set for each object.

These low-level features need to be mapped to intermediate semantic terms such as ‘circular’, ‘long’, ‘angular margin’, ‘spicular margin’, etc. These semantic terms, more familiar to an expert user, can be used to describe the class or category of an object in the image. This is the first phase of mapping to semantic terms.

To achieve fine-grained, high-level descriptions of objects is sometimes only possible by taking into account intermediate descriptors such as spatial information about location or relation to other objects

within the image. Coarse-grained descriptions of the complete scene require the grouping of objects and describing their spatial relation to each other. The intermediate level semantic terms need to be mapped to domain level terms which describe the content depicted in the image. This is the second phase of mapping to semantic terms.

There is no universal algorithm available that can automatically process the semantic information for all kinds of images. This means that specific images need special processing functions in order to implement the processing chain described in this section.

5. Results and Discussion

Table 1 shows the evaluation results of the neural network using the test data set of 1627 objects. User Labeled (User) is the number of objects in the test set that were labeled as a specific ‘class’ by the expert user. Objects Classified (ANN) is the number of objects in the test set that the network classified as ‘class’ while Objects Correctly Classified (OCC) is the number of the Objects Classified whose class label corresponds with that given by the expert user. Precision, Recall and f-measure are terms from the information retrieval domain [24]. Precision and recall are both measurements of classification quality. f-measure provides a more useful measure of the overall performance since it takes into account the generally opposing qualities of precision and recall (i.e. high recall generally results in lower precision and vice versa). They are calculated as follows:

- Precision (P) is Objects Correctly Classified / User Labeled
- Recall (R) is Objects Correctly Classified / Objects Classified
- f-measure (f-m) is $(2 \cdot R \cdot P) / (R + P)$

The accuracy of the system is calculated as the total number of correctly classified objects (1094) divided by the total number of objects (1627). This gives an accuracy rate of 0.627.

Table 1. Results of simulating the trained neural network using the test data set.

Class	User	ANN	OCC	P	R	f-m
ER	435	482	226	0.469	0.520	0.493
Go	696	928	474	0.511	0.681	0.584
Mi	38	30	30	1.000	0.789	0.882
MG	63	67	47	0.701	0.746	0.723
Ri	354	305	291	0.954	0.822	0.883
TV	41	28	26	0.929	0.634	0.754

The high precision for Mitochondria and to a lesser extent for Ribosome and Tubular Vesicles indicates the better visual distinction based around the shape of these objects – Mitochondria are large and tend to be

more circular (higher Eccentricity values); Ribosome are small and more irregular in shape while Tubular Vesicles are circular (very high Eccentricity values) and consistently very small in size.

The poor performance in identifying Endoplasmic Reticulum and Golgi objects is possibly due to their low visual distinction when only considering basic shape features. As section 4.1 discussed, they are much more easily described using texture and more intermediate descriptors such as spatial relations.

Neural networks are generally better at discriminating between classes, as is shown in [16]. Therefore the difficulty this network has in distinguishing between Go and ER objects is interesting. However, we feel this is attributable to the lack of spatial relations in the input features.

We didn't achieve the accuracy that we were hoping for since we mapped directly from the low-level features to the domain class label. In addition we did not have a large enough set of input features such as spatial relationship to other objects. This meant that information about spatial relations and intermediate shape and texture descriptors were not incorporated into the classification process.

Also the testing and training data all came from a single example cell. This means that this dataset might not represent adequate statistical variation among the data from different cells. It is not clear that this network would perform as well using data from another cell. When new data is added, the ANN needs to be retrained in order to achieve good performance. Neural nets do not support incremental learning. Other methods such as decision trees or case-based reasoning are preferred for this reason.

Overall, the small sample set and the limited variation in the source object (one cell) restricts the statistical significance of these results and the conclusions that can be made from this evaluation. However, the results from this network provide support for the view that mapping from low-level features to intermediate terms and then from intermediate terms to domain descriptions is likely to be a more successful approach.

6. Conclusion and Future Work

Using artificial neural networks to map from low-level media features directly to high-level semantic terms for image regions does not demonstrate a particularly high level of accuracy. While previous applications have shown that neural networks can be effective in image classification tasks [16,17,18], we believe that a multi-stage process, as proposed in

section 4.2, is likely to be more effective for semantic annotation of image regions. Building semantic annotations by mapping from low-level to visual semantic descriptors and then to domain semantic terms rather than mapping directly from low-level features (for example, as is the case in [11]) will enable richer and more accurate semantic annotation.

However, until efficient and accurate automatic segmentation algorithms are able to be developed, techniques are needed for semi-automatic semantic annotation that can handle small input data sets, evolving models and rapidly increasing data. Therefore, we aim to develop a system that can operate on an initial, small dataset but incrementally adapt and improve with the addition of further data as it becomes available. The classification system will eventually become more generalized and have improved accuracy as new cell slices are incorporated into the data set. This situation is common in many medical and scientific research fields where the available experimental data may initially be relatively small but which will increase as further experiments and analysis are conducted.

We believe that to generate a semantic description of an image you cannot use the low-level features directly, you have to first map them to intermediate symbolic or semantic terms that make sense for a domain expert. Therefore we intend to focus on a multi-step classification procedure where visual semantic terms (such as 'circular', 'fine speckled margin', 'adjacent' etc.) are created from the automatically extracted low-level features. These terms can then be used to build better classification systems using techniques such as inferencing rules [3,4], case-based reasoning [19,20] or decision trees [26].

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